Matthew Jeung

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Education

Northwestern University — Evanston, IL
PhD in Computer Science — *2026 -*

University of Chicago — Chicago, IL
BS in Computer Science, Specialization in Machine Learning — *2022 - 2026*

- **Thesis:** *Towards Programmable Matter: Design and Interaction for Modular Robotic Systems*

Honors and Awards

Computing Research Association Outstanding Undergraduate Researcher Honorable Mention	2025
UIST 2025 Best Demo Award (Jury's Choice)	2025
University of Chicago - Graduated with Honors in Computer Science	2026
University of Chicago - Cum Laude	2026
Quad Undergraduate Research Scholar Award ($5000)	2024-25
Jeff Metcalf Fellowship ($5000)	2023
Humanities UX Scholar	2022

Papers

[1] Shape n' Swarm: Hands-on, Shape-aware Generative Authoring with Swarm UI and LLMs.
M. Jeung, A. Sathya, M. Qian, S. Arellano, L. Jimenez, K. Nakagaki.
In Proc. UIST 2025. Acceptance Rate: 22.2%. **Best Demo Award.**

[2] CLAW: Learning Continuous Latent Action World Models via Adversarial Latent Regularization
T. Ayalew, S. Wheeler, **M. Jeung**, X. Zhang, A. Cruz Arce, K. Stocking, M. Maire, M. Walter.
Under Review at ICLR 2027.

[3] Paper under review at CHI 2027 (Anonymized)

[4] Paper under review at CHI 2027 (Anonymized)

Research Experience

Northwestern University - Center for Robotics and Biosystems — 2026 –
PhD Student — *Evanston, IL*

Toyota Technological Institute at Chicago — 2025 - 2026
Visiting Researcher at Robot Intelligence through Perception Lab (Advised by Prof. Matthew Walter) — *Chicago, IL*

- Led a modular robotic locomotion project using deep reinforcement learning.
- Engineered a novel cube robot with an imbalanced center of mass for full 3D reorientation using a single actuator.
- Contributed to a deep learning project on unsupervised action representation learning.
- Developed PyTorch code for MuJoCo simulation, GPU training, and benchmark integration.

University of Chicago - Actuated Experience Lab — 2023 – 2026
Undergraduate Researcher (Advised by Prof. Ken Nakagaki) — *Chicago, IL*

- Introduced a new hand and speech-based approach for generative control of tabletop swarm robots.
- Engineered a novel modular haptic device and system, in industry collaboration with Sony.

Coursework

Theory of Algorithms; Math Foundations for Machine Learning; Engineering Interactive Devices; Mobile Computing; Introduction to Designing Interaction; Engineering Interactive Electronics onto Printed Circuit Boards; Introduction to Machine Learning; Robot Planning, Learning, and Estimation; Generative Deep Models; Foundations of Machine Learning - Generative AI; Swarms and Multi-Robot Systems

Teaching

Toyota Technological Institute at Chicago 2025
Robot Arm Course Instructor *Chicago, IL*

- Mentored five HS students through hands-on robot manipulation course as part of a TTIC outreach program.
- Developed computer vision curriculum and project to automate robot arm playing TicTacToe.
- Gave lectures on python and computer vision.

Volunteering and Activities

Phoenix Sustainability Initiative 2023 – 2024
Green Data Team *Chicago, IL*

- Performed Python data analysis and visualization on Chicago environmental justice and health conditions.

University of Chicago Asian American Intervarsity 2024 - 2026
Service Coordinator *Chicago, IL*

- Organized and participated in regular tutoring, care-packaging nights with local organizations.
- Developed partnerships with local service providers in Chinatown and South Chicago.

UChicago Robotics Team 2024 - 2025
Combat Robotics Team *Chicago, IL*

- Won MakeHarvard 2025 Most Sustainable Project with glove integrated with grocery sustainability feedback
- Designed 1-lb combat robot from scratch, developing Fusion 360, electronics, and prototyping skills

Technical Skills

Programming: Python, C, C++, Java, Swift, JavaScript, PyTorch
Electronics and Hardware: Microcontrollers (Arduino and Raspberry Pi), Circuit Board Design (KiCad), CAD Design (Fusion 360), Soldering, Laser Cutting
Design: Figma, Adobe Photoshop, Final Cut Pro